Executive Summary

This report presents the project plan for the Student Engagement & Rewards Module, Gamification Team, as part of the CPS 714 Software Project Management course. The project demonstrates the **core application of project management principles,** given a **two-month duration** and a project budget of **$160,000**.

Key deliverables for this assignment include Business Case, Project Charter, Scope Statement, Work Breakdown Structure (WBS), Gantt Charts, A Network Diagram, and Resource Assignment Breakdown. All together, these documents outline the module's objectives, approach, risks, timeline, and success criteria.

This assignment provides a well-structured foundation for project execution, ensuring that all objectives can be delivered within scope, time, and budget constraints.

Assignment Tasks

1. Business Case

Prepared by**:** Jackie Li

Date: 10/12/2025

Introduction

With the return of in-person classes over recent years, academic institutions are increasingly focusing on enhancing student engagement for on-campus events and programs. Many of these initiatives rely on manual tracking and data input, resulting in inconsistent participation data and making it difficult to reward active students effectively.

The **Student Engagement and Rewards Module** aims to gamify student engagement by rewarding points for participation in events and allowing students to redeem their points through a digital rewards catalogue. This project forms part of the broader CampusConnect initiative to modernize student services and foster community through technology.

Objective

The objective of the **Student Engagement and Rewards Module** is to **increase student participation and engagement** by implementing a gamified system that rewards student involvement across university events and programs.

Situation, Problem & Opportunity Statement

Currently, student engagement data is isolated across many platforms and parties; as a result, event and program organizers often **manually compile the data**. Event attendance is typically recorded via sign-in sheets or online attendance forms, resulting in data inaccuracies and limited insights into participation patterns.

The **absence of a centralized tracking system** leads to inaccurate, inefficient, and low insight visibility for both parties. Ultimately, this **reduces the motivation** for students to attend events or programs.

By introducing and implementing a **gamified** concept via a **points-based rewards system and a centralized leaderboard**, the university can not only motivate students to further engage in said activities but also **gain actionable analytics** on student engagement trends across faculties.

Critical Assumption and Constraints

The following consists of a list of assumptions:

- Student will automatically opt into the rewards system via their CampusOne accounts

The following consists of a list of constraints:

- Fixed budget of $160,000
- Total project duration of two months

Analysis of Options and Recommendation

Option	Description	Advantage	Disadvantage
Do Nothing	Allow event and program organizers to continue with manual tracking	No upfront cost	Continue with ongoing inefficiencies; no engagement incentive for students.
Purchase Existing Third-Party Solutions	Purchase a third-party engagement platform	Easy and fast deployment	Costly licensing, limited customizations and integration with CampusConnect
Develop In-House	Implement a gamified	Full customization,	Requires internal

Model	system in CampusConnect	integration flexibility, and easy data control.	development resources and system maintenance.

Table 1: Analysis Options

Proceed with the development of the **in-house module**. This option provides the greatest alignment with institutional goals, allowing easy integration with the existing CampusConnect system. The option also offers long-term scalability for future engagement initiatives.

Preliminary Project Requirements

The following is a list of requirements for the project.

- **Points** - Points system to track and reward event and program attendance
- **Rewards Catalog** - The ability to redeem items (e.g. campus merch and gift cards)
    - **Admin** - Allow staff to manage rewards catalogue
- **Leaderboard** - Display highly engaged students. Based on engagement and not points

Budget Estimate and Financial Analysis

With a fixed budget of **$160,000,** the allocation of the budget can be seen in the following:

Category	Cost Estimate
Man Power (Personnel)	$90,000
Software & Licensing	$15,000
Training & Documentation	$10,000
Testing & QA	$8,000
Cloud Infra & Hosting	$20,000
Contingency	$17,000
Total	$160,000

Table 2: Budget Estimate Breakdown

With the reduction of administrative workload and higher engagement of students, the long-term cost savings will be equivalent to $50,000 annually in staff hours and management overhead.

Schedule Estimate

The development schedule is estimated to span a total of 7 weeks.

Phase	Duration Week(s)	Timeline
Project Initiation & Requirement Solicitation	0.4 Weeks = 3 Days	Oct 26 - Oct 28
Design & Documentation	0.86 Weeks = 6 Days	Oct 29 - Nov 4
Development	2 Weeks	Nov 5 - Nov 19
Testing & QA	0.4 Weeks = 3 Days	Nov 18 - Nov 21
Integration	1 Week	Nov 24 - Nov Dec 1

Table 3: Development Schedule Estimate

Potential Risks

Risk	Impact	Mitigation
Integration Challenges	High	Early review with other teams to ensure API testing works.
Limited Student Adoption	Medium	Marketing campaign and pilot testing with student clubs.
Schedule Delays	Medium	Usage of Agile sprints and weekly progress reviews
Budget Overrun	Low	Maintain a 10% contingency fund and track the budget weekly.

Table 4: Project Risk Breakdown

Exhibits

Year	Future Benefit	Discount Factor (6%)	Present Value
1	$60,000	0.943	$56,580

2	$60,000	0.890	$53,400
3	$60,000	0.840	$50,400
4	$60,000	0.792	$47,520
5	$60,000	0.747	$47,520
Total Present Value	-	-	$252,720

Table 5: Exhibit A - Financial Analysis [1]

Category	Amount
Project Cost	$160,000
Total Present Value	$252,720
Net Present Value	$92,720
Payback Period	~2.8 Years

Table 6: Exhibit A - Payback Period Calculation

2. Project Charter

Prepared by**:** Clayton Barnett, Alejandro Medina Eraso

Date: 10/7/2025

Project Start Date	
Project Finish Date	
Project Manager	Jackie Li - jackie1.li@torontomu.ca

Table 7: Project Charter Breakdown

Objectives

Implement the reward and gamification features of the CampusConnect App. Integrate with the User Dashboard and Event tracking teams to ensure proper functioning of the features within the overall app.

Success Criteria

The project should finish at least several days before the finish of the overall CampusConnect development project. All the written requirements should be implemented and thoroughly tested. The result should improve the app's user experience.

Approach

- Within two weeks, devise a clear scope document describing the requirements and deliverables. This will be used to create a clear Work Breakdown Structure and GANTT chart, outlining the specific tasks that will be required and who will do them.
- Within two weeks, create a change request and a change control strategy to ensure alignment
- Within two weeks, create a time management strategy by identifying milestones and task durations
- Use the GANTT chart and change control strategy to complete the tasks according to the schedule
- Involve the entire 5-person team in the planning and implementation phases of the project.
- Hold weekly meetings to ensure progress is moving steadily
- The project manager will meet weekly with the other CampusConnect Project managers to ensure alignment across the larger project
- Use GitHub according to the strategy outlined by the Project managers

Roles & Responsibilities

Name and Signature	***Role***	***Position***	***Contact Information***
Clayton Barnett - CB	Team Member	Developer	clayton.barnett@torontomu.ca
Ansugan Subramaniam	Team Member	Developer	ansugan.subramaniam@torontomu.ca
Omar Ahmed	Team Member	Developer	omar4.ahmed@torontomu.ca
Alejandro Medina	Team Member	Developer	alejandro.medina@torontomu.ca
Jackie Li - JL	Team Lead	Developer	jackie1.li@torontomu.ca

Table 8: Roles & Responsibilities Contact Information

Comments

3. Change Request

Prepared by**:** Omar Ahmed, Ansugan Subramaniam

Date: 10/7/2025

Date Request Submitted	Oct 1, 2025
Title of Change Request	Update to Rewards System
Change Order Number	01
Submitted By	TMU Events Coordinator
Change Category	• Scope • Schedule • Cost • Technology • Other

Table 9: Change Request Breakdown

Description

The module awards students points only for completing assignments and attending classes. The proposed change is an enhancement to include reward points for participation in extracurricular activities and volunteering events. The enhancement aims to introduce new functionalities that enable the institution to record, manage, and reward extracurricular engagement through a structured point system.

Events Prompting Change

Since the beginning of the school year, the TMU Events Board saw a 30% decline in student participation at extracurricular events. This could be due to students focusing only on academics because of the new rewards system. This problem caused a decrease in the school's revenue, decreased student satisfaction, and could impact the school's reputation in the future.

Justification

This change is necessary because it will encourage students to participate in university activities beyond academics. By creating a new incentive for students to participate in community

involvement, students will participate in more events, increasing ticket revenue, strengthening student engagement and school spirit, and improving students' overall mental health.

Impact Assessment

- **Scope:** The project scope will be expanded to include new functional areas within the Student Engagement and Rewards Module.
- **Schedule:** The inclusion of these new features is expected to add approximately 2–3 weeks to the current project timeline.
- **Cost:** The proposed change will result in an estimated cost increase of $2,500–$3,000
- **Staffing:** The change will require involvement from a backend developer, a frontend developer, and a QA tester.
- **Risk:** Errors in event verification or point assignment could lead to disputes or inaccurate scores. Also, adding too many activity types without clear rules may complicate the system.
- **Other:** The point system and participation criteria should be reviewed to ensure fairness and transparency.

Implementation Plan

1. Update the database to include new event and volunteer activity categories.
2. Design UI updates for "Rewards" dashboard.
3. Implement event check-in with the Student IDs.
4. Adjust the reward calculation algorithm.
5. Conduct user acceptance testing with a small student group.
6. Roll out an update during mid-semester to maximize engagement for upcoming events.
7. Market the new feature through social media and student emails

Required Approvals

Name/Title	**Date**	**Approve/Reject**
Project Sponsor - TMU Student Engagement Director		
Project Manager - CampusConnect Development Team		
Finance Officer - TMU Students Affairs		
QA Lead - CampusConnect		

Table 10: Approvals Signage

Scope Management

4. Requirements Collection and Traceability Matrix

Prepared by: Clayton Barnett

Date: 10/22/2025

Requirements Elicitation Process

The strategy we will use to elicit requirements is to interview and/or survey members from five distinct CampusConnect stakeholder groups. We will aim to get one or two requirements from each stakeholder. These groups are:

1. Members of the Core Services team (Team 3): This team is in charge of providing event organizers with ways of tracking the attendance of their events through either a check-in list or a QR code. We will need to communicate with this team to get information on how exactly they will implement the attendance tracking, so that we can use this information properly.

2. Members of the User Engagement Team (Team 4): This team is in charge of the user's personal dashboard. This dashboard will display points earned and allow students to redeem those points for rewards. We will need to communicate with this team to get information on how to integrate the points system into their display, including an optional overall points leaderboard.

3. Campus/local shops: These will be the providers of rewards. We will need to communicate with them to see what awards they can provide, and how they would like students to redeem them - for instance, through a QR code or a promo code.

4. Members of the Comms team (Team 8): This team is in charge of communications with students, and in particular about reward opportunities. We will need to speak with them to understand how we can integrate rewards into their notifications.

5. Student Event Organizers: There are two subcategories within this stakeholder group: TMU staff and administrators who organize student events, and students themselves who are involved in student groups that organize events. We will need to speak with members from both subgroups to understand what sort of incentive system they would like to see for their events.

6. General TMU students: These will be the end users of the application. We should speak to them to see what aspects of the gamification are most appealing to them - for example, a leaderboard, different types of rewards, or otherwise.

These stakeholders will require different communication approaches to elicit the requirements that are important to them:

- For members of the CampusConnect team, a direct in-person or virtual interview can be conducted.
- For local and campus shops, as well as student event organizers, communication will happen via email.
- For general TMU students, an online survey will be disseminated in a number of different classes and departments.

Requirements Traceability Matrix

Req. No.	**Name and Description**	**Category**	**Source**	**Status**
1	*Student Attendance and Points Data Integration* - Each student should have a data entry that can track both their event attendance and their points, which could enable automatic point assignment	Data Management	Core Services Team and User Engagement Team	Not implemented.
2	*Event Priority Assignment* - Organizers should have a way to assign an event a high or low priority, which helps to determine how many points students receive for attending	User Features	Event Organizers	Not implemented.

3	*Shop Rewards Management* - Shops must have a way to add rewards and validate reward codes.	User Features	Local and Campus Shop Owners	Not implemented.
4	*New Rewards Notification* - When a new reward is added to the catalogue, the Comms team should receive a notification with details about the reward, so they can include it in their email notifications.	System Notifications	Comms Team	Not implemented.
5	*Reward Sharing* - Students would like to have a way to give rewards to others if they don't want to use it themselves.	User Features	TMU Students	Not implemented.

Table 11: Traceability Matrix

5. Project Scope Statement

Prepared by**:** Ansugan Subramaniam

Date: 10/22/2025

Project Justification

The Student Engagement and Rewards Module is proposed to address the growing need for improving student motivation, participation, and retention within the institution. Current engagement levels suggest that students often lack incentives to participate actively in academic and extracurricular activities. By introducing a digital rewards system that recognizes and celebrates student achievements, the institution can foster a more engaging and motivating learning environment.

Product Characteristics and Requirements

1. It should record and track student activities (attendance, submissions, participation).
2. It should award badges or rewards automatically when the criteria are met
3. The interface must be intuitive and accessible for users with varying technical skills.
4. The system should allow easy updates and additions of new reward types or engagement criteria.
5. It must protect sensitive student information using secure login and data encryption.

Product User Acceptance Criteria

The Student Engagement and Rewards Module will be accepted once it successfully tracks student activities, awards points and badges automatically, and integrates seamlessly with the existing system. The system must be user-friendly, secure, and responsive across all devices. All data must be accurate, protected, and updated in real-time. Final acceptance will be granted after the project sponsor, project manager, academic administration, and IT department verify that all functional, performance, and security requirements have been met.

Summary of Project Deliverables

Project Management-Related Deliverables:

- **Project Charter** – Defines the project objectives, scope, stakeholders, and overall authorization to begin the project.
- **Gantt Chart** – Displays all project tasks, timelines, dependencies, and milestones.
- **Risk Register** – Identifies any risks, their impact, probability, and strategies
- **Change Request Log** – Documents any proposed changes to scope, schedule, or cost, along with approval status.

Product-Related Deliverables:

- **Student Dashboard** – Displays points, badges, rewards, and leaderboard standings in real-time.
- **User Manual and Training Guide** – Step-by-step instructions for students, faculty, and administrators on how to use the system.
- **Developed Module** – Fully functional Student Engagement and Rewards Module, integrated with the institution's system
- **Testing and Quality Assurance Report** – Documentation of all testing phases, including system, integration, and user acceptance tests.

6. Work Breakdown Structure

Prepared by: Alejandro Medina Eraso

Date: 10/21/2025

1.0 Points System Development
- 1.1 Requirements Analysis
  - 1.1.1 Define point system rules and participation categories
  - 1.1.2 Design SQL schema and data flow diagram
- 1.2 Backend Implementation
  - 1.2.1 Develop Node.js allocation algorithm
  - 1.2.2 Create RESTful APIs for point operations
- 1.3 Testing and Validation
  - 1.3.1 Unit test

2.0 Rewards Catalog Development
- 2.1 Requirements and Interface Design
  - 2.1.1 Define redemption rules and reward categories
  - 2.1.2 Design catalog UI
- 2.2 Backend and Database Development
  - 2.2.1 Build SQL reward tables and relationships
  - 2.2.2 Implement Node.js APIs for reward redemption
- 2.3 Testing and Quality Assurance
  - 2.3.1 Perform functional testing and reward flows
  - 2.3.2 Fix and verify identified errors

3.0 Leaderboard Feature Development
- 3.1 Metrics and Ranking Logic
  - 3.1.1 Define ranking criteria and tie-breaking rules
  - 3.1.2 Develop Node.js ranking algorithm with SQL queries
- 3.2 Frontend and Integration
  - 3.2.1 Create leaderboard UI
  - 3.2.2 Integrate live ranking data via API endpoints
- 3.3 Testing and optimization
  - 3.3.1 Validate data accuracy and refresh timing

4.0 System Integration, Testing and Deployment
- 4.1 Cross-Module Integration
  - 4.1.1 Connect gamification APIs with the dashboard and notifications

4.1.2 Verify data synchronization between subsystems
4.2 Deployment and Environment Setup
4.2.1 Containerize services using Docker
4.2.2 Deploy finished product

Dictionary Entries

1.0 Point System Development

WBS Item Number: 1.1.1
WBS Item Name: Define point system rules and participation categories
Description:
This task involves determining how students will earn and lose points through various activities and events. The team will define participation categories such as academic, social, and athletic events, each with its own point value. Bonus conditions (e.g., volunteering, event hosting) and special cases (e.g., event cancellations or attendance conflicts) will also be taken into account. These rules ensure fairness and consistency across the platform.

WBS Item Number: 1.1.2
WBS Item Name: Design SQL schema and data flow diagram
Description:
A relational database schema will be created to define how points, users, and events are stored and related. The team will design ER diagrams illustrating table relationships. Data flow diagrams will map interactions between the Event Management System, backend APIs, and the SQL database. The design will support real-time updates and scalability for thousands of users.

WBS Item Number: 1.2.1
WBS Item Name: Develop Node.js allocation algorithm
Description:
This step involves implementing backend business logic using Node.js to calculate and assign points to users based on event attendance data. The algorithm will include validation layers, error handling, and rollback mechanisms to maintain data consistency. It will also manage bonus allocations.

WBS Item Number: 1.2.2
WBS Item Name: Create RESTful APIs for point operations
Description:
APIs will be developed using Node.js to enable CRUD operations on point records. These APIs will allow the frontend and other modules to retrieve, add, or modify user point balances securely.

WBS Item Number: 1.3.1
WBS Item Name: Unit test
Description:
Unit tests will be written to confirm that point calculations match the rules established in the requirements analysis. Tests will include attendance and invalid attendance data, duplicate submissions and bonus event scenarios.

2.0 Rewards Catalog Development

WBS Item Number: 2.1.1
WBS Item Name: Define redemption rules and reward categories
Description:
This task will define the types of awards available to participants, including physical or digital items to be won. The team will establish redemption criteria, frequency limits and reward expiry dates. Rules will also be established in order to ensure equal reward distribution and prevent exploitation.

WBS Item Number: 2.1.2
WBS Item Name: Design catalog UI
Description:
The frontend team will design a user interface, creating mockups that would illustrate the browsing, filtering, and redemption flows, ensuring a design that functions properly across devices and platforms.

WBS Item Number: 2.2.1
WBS Item Name: Build SQL reward tables and relationships
Description:
Create database structures for storing reward data, redemption history, and user transactions on SQL. Relationships between users, rewards and point balances will be clearly defined. Constraints and foreign keys will be determined upon creation.

WBS Item Number: 2.2.2
WBS Item Name: Implement Node.js APIs for reward redemption
Description:
Backend endpoints will be developed to process reward redemption requests. Logic will verify reward availability, record transactions, and deduct points if redemption is successful. Error handling will prevent duplicate redemption attempts and rollback failed operations. APIs will also send confirmation responses to the user dashboard.

WBS Item Number: 2.3.1
WBS Item Name: Perform functional testing of reward flows
Description:
This task will ensure end-to-end functionality, from browsing to reward redemptions. Tests will ensure consistency and reliability, validating UI behaviour, API responses, and database updates.

WBS Item Number: 2.3.2
WBS Item Name: Fix and verify identified errors
Description:
Any errors discovered during testing will be recorded and fixed, with each resolved error being tested.

3.0 Leaderboard Feature Development

WBS Item Number: 3.1.1
WBS Item Name: Define ranking criteria and tie-breaking rules
Description:
This task will establish the logic for leaderboard rankings, determining how points are collected and how ties should be resolved. The design will include criteria such as cumulative points, types of events and the number of events attended.

WBS Item Number: 3.1.2
WBS Item Name: Develop Node.js ranking algorithm with SQL queries
Description:
Backend logic will be developed to determine rankings using SQL. The algorithm will fetch, sort and cache data to ensure real-time updates with minimal latency.

WBS Item Number: 3.2.1
WBS Item Name: Create leaderboard UI
Description:
A leaderboard interface will be designed and built using React. The UI design implementation will be determined during the creation process.

WBS Item Number: 3.2.2
WBS Item Name: Integrate live ranking data via API endpoints
Description:
The leaderboard interface will be connected to backend APIs for real-time updates. The frontend will fetch data periodically and update the interface without requiring a page refresh.

WBS Item Number: 3.3.1

WBS Item Name: Validate data accuracy and refresh timing
Description:
Tests will confirm that the leaderboard interface updates accurately. By simulating events, we can test ranking recalculations, data consistency, and how often the interface refreshes to update the leaderboard.

4.0 System Integration, Testing and Deployment

WBS Item Number: 4.1.1
WBS Item Name: Connect gamification APIs with dashboard and notifications
Description:
Connect the gamification APIs to the main CampusConnect dashboard and notification subsystems. The connection will allow users to view their points, rewards, and leaderboard status.

WBS Item Number: 4.1.2
WBS Item Name: Verify data synchronization between subsystems
Description:
Testing will be used to confirm that data flows consistently within points, rewards and leaderboard modules.

WBS Item Number: 4.2.1
WBS Item Name: Containerize services using Docker
Description:
Dockerfiles will be written for backend, frontend, and database components, using Docker to define how these containers will interact locally.

WBS Item Number: 4.2.2
WBS Item Name: Deploy finished product
Description:
The final task is to deploy the finished product, deploying the containerized services to the production server or cloud environment, and monitoring for any possible errors that show up afterwards.

7. Gantt Chart (Version 1)

Prepared by**:** Jackie Li

Date: 10/20/2025

Initial Gantt chat with activities based on the Work Breakdown Structure (WBS).

	Task Mode	Task Name	Duration	Start	Finish
1		**Point System Phase**			
2		**Requirements Analysis**			
3		System Rules			
4		SQL Schema			
5		Data Flow Diagram			
6		**Backend Implementation**			
7		Project Architecture Development			
8		RESTful Endpoints			
9		**Testing & Validation**			
10		Business Logic			
11		Public Endpoints			
12		**Rewards Catalog Phase**			
13		**Requirements & Interface Design**			
14		Define Redemption Rules			
15		Design UI Catalog			
16		**Backend & Rewards Database Design**			
17		Design SQL Rewards Tables			
18		Implement Redemption Logic			
19		Create RESTful API			
20		**Testing & QA**			
21		Perform Functional Tests			
22		**Leaderboard Phase**			
23		**Metric & Ranking Logic**			
24		Ranking & Tie-Breakers			
25		Develop Ranking Backend			

Figure 1: Gantt Chart v1[2]

26		**Frontend Leaderboard**			
27		Create Leaderboard UI			
28		Integrate Live Data Feed			
29		**Testing & Optimization**			
30		Validate Leaderboard Accuracy			
31		Optimize Leaderboard Performance			
32		**System Integration & Deployment Phase**			
33		**Cross-Module Integration**			
34		Gamification Connection			
35		Verify Data Synchronization			
36		**Development & Environment Setup**			
37		Containerize Service via Docker			
38		Deploy Module to Production			

Figure 2: Gantt Chart v1[2]

8. Scope Validation & Change Control Strategy

Prepared by**:** Omar Ahmed

Date: 10/21/2025

Scope Validation Strategy

- Scope validation for the Student Engagement and Rewards Module will be accomplished via regular review meetings with stakeholders, which include the Project Sponsor, IT Department and Academic Administrators.
- Every feature, including the Student Dashboard, Rewards Module, and User Manual, will go through structured validation testing to ensure that it meets the requirements, user acceptance criteria, and integration standards.
- Formal sign-off will be required at the completion of each major milestone, moving on to the next phase.

Change Control Strategy

- Change control will follow a structured process using the project's Change Request Log. This log allows for documentation, evaluation, and approval or rejection of the proposed changes.
- Every change request will be assessed based on its impact on the scope, schedule, cost and quality.
- Approved changes will require an update to the project plan and Gantt chart, as well as approval from the stakeholders, before implementation.

These strategies ensure that every potential project modification is managed properly, with minimum disruption to the project's progress while also maintaining the project's overall goals.

Time Management

Prepared by: Alejandro Medina Eraso

Date: 10/23/2025

9. New Activities Proposal

Determining an accurate estimation of resources, costs, and durations is essential to the success of the CampusConnect Project. While the Gamification Module team's focus is on developing a points system, rewards catalog, and leaderboard, additional activities that go beyond the immediate scope can provide a significant improvement in project estimates. These activities will be able to greatly strengthen the team's understanding of financial, managerial, and operational factors that affect project execution, ensuring that the planning process of the project is not only technically sound but also feasible in terms of finances, resources, and duration.

One possible activity involves **meeting with a software development accountant to refine the project's cost estimation process**. This consultation would be beneficial in helping the team understand the financial needs of the project by discussing budgeting strategies, cloud hosting fees, API usage fees and developer hourly rates. The accountant's input would ensure that the project stays within budget and help with identifying potential cost-saving strategies to keep the project sustainable long-term, giving the team the ability to provide a realistic estimate of the resources required. This activity can be made into a task, a **3rd Party Cost Analysis,** and added to the Gantt chart.

To further improve duration and resource estimation capabilities, the team can **review past project documentation from similar projects**. Analyzing past Gantt charts, resource usage reports, and durations can provide data for task durations and common bottlenecks to avoid. These insights can be used as a reference for current duration and resource estimates. By understanding the concrete data from previous projects, the team can create a more accurate resource and time estimation report supported by evidence-based research. Another activity that can benefit the project would be to **interview an experienced project manager who has led similar software development projects.** This would provide information that would help the team identify useful patterns in time management, scheduling, and task prioritization, and how to apply them to our own project. The project manager could share valuable lessons learned, possible pitfalls to avoid, how to effectively handle delays and strategies for managing teams. This would help the team determine its own duration estimates based on the first-hand experience of the project manager in managing similar projects. These two activities can be

grouped into a single overarching task, **Organizational Alignment**, that can be added to the Gantt chart.

Another activity that can benefit the project would be to ensure the proposed project addresses genuine student needs and delivers meaningful value to the institutions. A **market validation study will be conducted through targeted student surveys and focused groups.** These activities will gather feedback on the current proposal and desired features, usability expectations, and current pain points with existing solutions. The analysis would help identify priorities, needs, and validate the overall relevance of the project before significant development resources are committed. This activity can be summarized in the task **Market Viability,** which can be added to the Gantt chart.

Together, these proposed activities extend beyond the immediate scope of the Gamification Module team, but contribute immensely to the estimation and planning of the project. Each activity provides a unique approach to improving the quality and reliability of the estimates of time, cost, and resources. Through expert consultations, student surveys, and analysis of historical data, the Gamification Module team can provide a more accurate, evidence-based estimate of cost, duration, and resource needs, increasing the likelihood of a successful project.

10. Milestone Identification

Prepared by**:** Omar Ahmed

Date: 10/21/2025

Milestone **Estimated Completion Date**

Requirements Finalization **Week 2**
Formal authorization of all project requirements from stakeholders. This milestone is specific (confirming the documented requirements for all modules) and measurable through the receipt of approval from all five stakeholders. It is time-bound, with completion targeted by the end of Week 2.

Database Schema and Data Flow Approval **Week 3**
Once the points system and rewards SQL schema and detailed data flow diagram are completed, the data development team will conduct a structured review to confirm compliance with the requirements. The milestone is measurable when the schema is validated and approved in peer review with no critical issues outstanding. It is achievable and relevant as it builds directly on the System Rules, SQL Schema, and Data Flow Diagram tasks.

Backend Core Completion **Week 4**
The database schema, backend business logic, rewards logic, and RESTful API are finalized and tested. This milestone is specific and measurable through successful code review and a minimum of 90 percent unit test coverage, and is time-bound to the end of Week 4.

Frontend Integration **Week 4**
Occurs in conjunction with Backend Core Completion. The frontend user interfaces are connected to the backend via RESTful APIs. This milestone is specific to the Create Leaderboard UI and Frontend Integration tasks and measurable through successful integration testing of all primary user interactions.

Module Testing and Validation Complete **Week 5**
User dashboards and administrative panels are connected to the backend. This milestone is measurable through usability and accessibility testing to confirm interface performance and responsiveness, with completion defined as all test cases passing and fewer than five percent defects remaining. It is relevant to ensuring compliance with design and accessibility standards.

System Integration & Synchronization Testing **Week 5**
The system is connected with TMU's student information and events management platforms. This milestone is specific to the Verify Data Synchronization and System Integration and Synchronization Verified tasks and is measurable through end-to-end synchronization testing that yields no data mismatches. It is relevant and achievable as it ensures compatibility with existing institutional infrastructure.

Deployment Readiness **Week 6**
All documentation, Docker containers, and system configurations are finalized for production deployment. This milestone is measurable through a completed deployment checklist and confirmation that all pre-deployment issues are resolved. It is time-bound, marking readiness before the start of Week 6 deployment.

Project Completion **Week 6**
The module is deployed to production, and final acceptance is obtained from all stakeholders. This milestone is specific to the Deploy Module to Production and Project Completion tasks, measurable through verified deployment logs and satisfied acceptance criteria, relevant as it marks delivery of the project's final objective, and time-bound to the end of Week 6.

11. Updated Gantt Chart (Version 2)

Prepared by**:** Jackie Li

Date: 10/23/2025

This version of the Gantt chart contains the new activities and milestones, but does not yet include the durations and dependencies.

	Task Mode	Task Name	Duration	Start	Finish
1		Organization Alignment			
2		Market Viability			
3		3rd Party Cost Analysis			
4		Requirements Finalization			
5		Database Schema & Data Flow Approval			
6		**Point System Phase**			
7		**Requirements Analysis**			
8		System Rules			
9		SQL Schema			
10		Data Flow Diagram			
11		**Backend Implementation**			
12		Project Architecture Development			
13		RESTful Endpoints			
14		**Testing & Validation**			
15		Business Logic			
16		Public Endpoints			
17		**Rewards Catalog Phase**			
18		**Requirements & Interface Design**			
19		Define Redemption Rules			
20		Design UI Catalog			
21		**Backend & Rewards Database Design**			
22		Design SQL Rewards Tables			
23		Implement Redemption Logic			
24		Create RESTful API			
25		**Testing & QA**			

Figure 3: Gantt Chart v2, Milestones & New Activities [3]

26			Perform Functional Tests			
27			Backend Core Completion			
28			Frontend Integration			
29			**Leaderboard Phase**			
30			**Metric & Ranking Logic**			
31			Ranking & Tie-Breakers			
32			Develop Ranking Backend			
33			**Frontend Leaderboard**			
34			Create Leaderboard UI			
35			Integrate Live Data Feed			
36			**Testing & Optimization**			
37			Validate Leaderboard Accuracy			
38			Optimize Leaderboard Performance			
39			Module Testing & Validation Complete			
40			System Integration & Synchronization Verified			
41			Deployment Readiness			
42			**System Integration & Deployment Phase**			
43			**Cross-Module Integration**			
44			Gamification Connection			
45			Verify Data Synchronization			
46			**Development & Environment Setup**			
47			Containerize Service via Docker			
48			Deploy Module to Production			
49			Project Completion			

Figure 4: Gantt Chart v2, Milestones & New Activities [3]

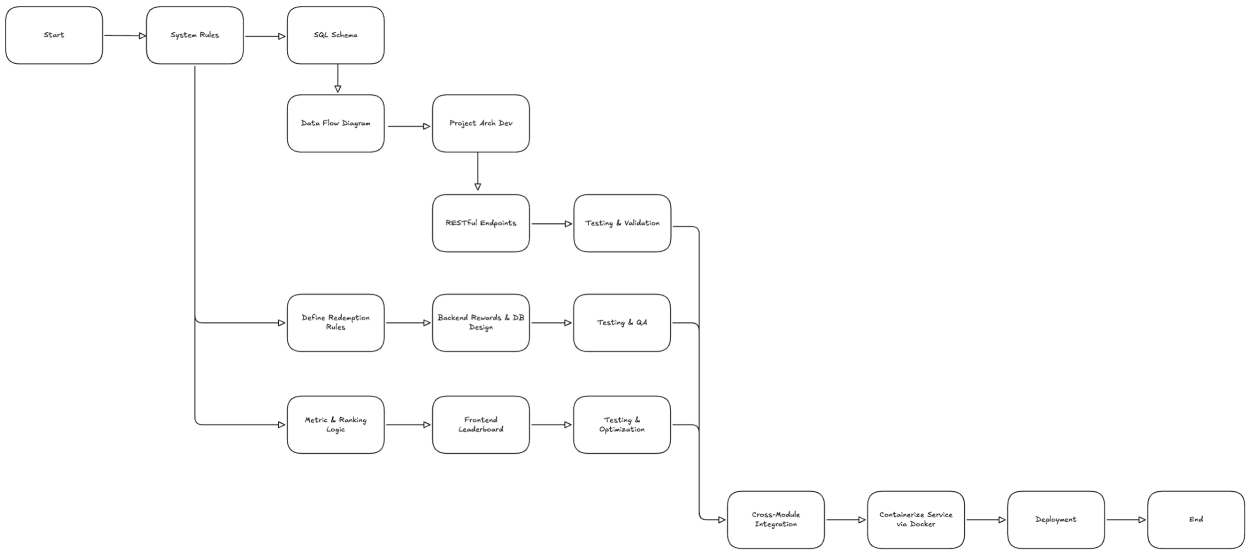


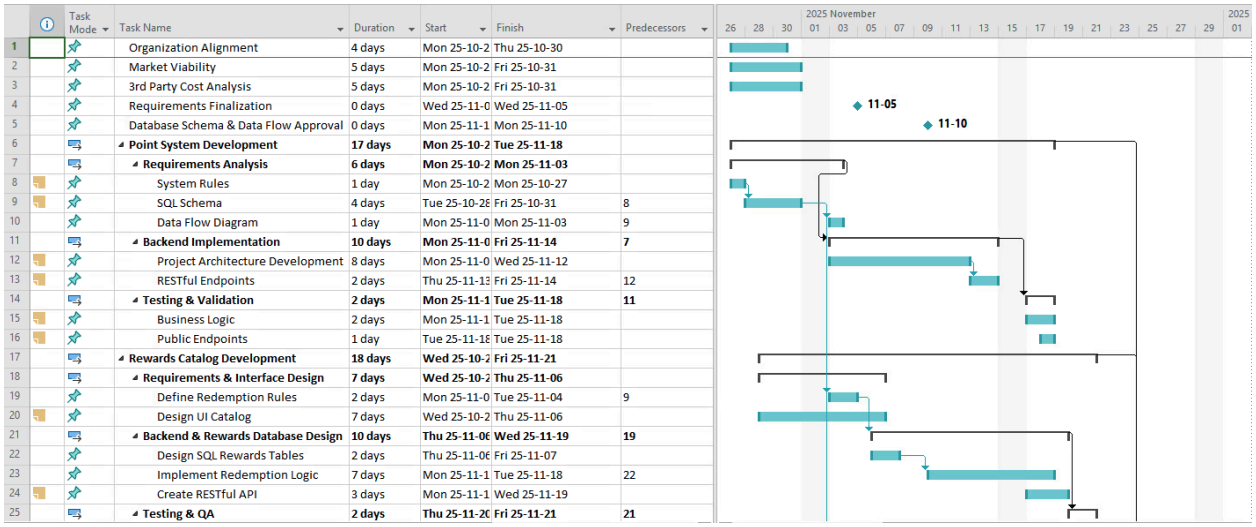
Figure 5: AON Network Diagram [4]

12. Task Durations and Dependencies (Gantt Chart Version 3)

Prepared by: Jackie Li

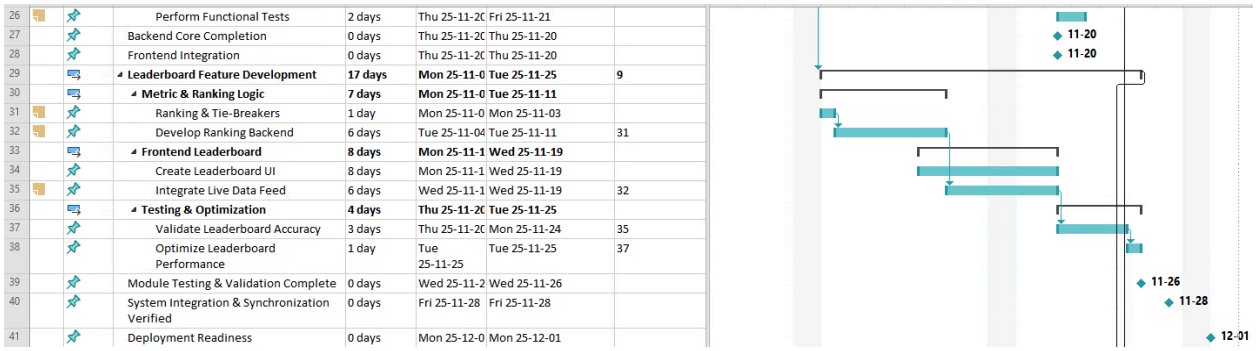
Date: 10/24/2025

Version 3 of the Gantt chart improves version 2 with the inclusion of task durations and dependencies.



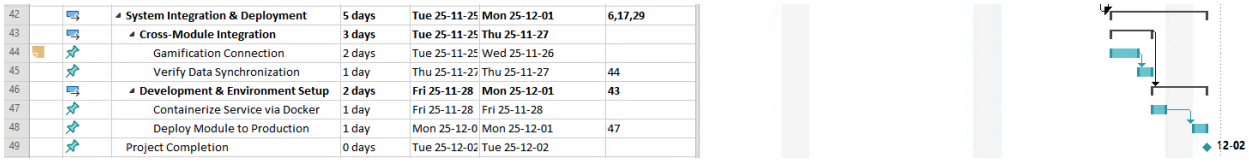
	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		Organization Alignment	4 days	Mon 25-10-2	Thu 25-10-30	
2		Market Viability	5 days	Mon 25-10-2	Fri 25-10-31	
3		3rd Party Cost Analysis	5 days	Mon 25-10-2	Fri 25-10-31	
4		Requirements Finalization	0 days	Wed 25-11-0	Wed 25-11-05	
5		Database Schema & Data Flow Approval	0 days	Mon 25-11-1	Mon 25-11-10	
6		**Point System Development**	**17 days**	**Mon 25-10-2**	**Tue 25-11-18**	
7		**Requirements Analysis**	**6 days**	**Mon 25-10-2**	**Mon 25-11-03**	
8		System Rules	1 day	Mon 25-10-2	Mon 25-10-27	
9		SQL Schema	4 days	Tue 25-10-28	Fri 25-10-31	8
10		Data Flow Diagram	1 day	Mon 25-11-0	Mon 25-11-03	9
11		**Backend Implementation**	**10 days**	**Mon 25-11-0**	**Fri 25-11-14**	**7**
12		Project Architecture Development	8 days	Mon 25-11-0	Wed 25-11-12	
13		RESTful Endpoints	2 days	Thu 25-11-13	Fri 25-11-14	12
14		**Testing & Validation**	**2 days**	**Mon 25-11-1**	**Tue 25-11-18**	**11**
15		Business Logic	2 days	Mon 25-11-1	Tue 25-11-18	
16		Public Endpoints	1 day	Tue 25-11-18	Tue 25-11-18	
17		**Rewards Catalog Development**	**18 days**	**Wed 25-10-2**	**Fri 25-11-21**	
18		**Requirements & Interface Design**	**7 days**	**Wed 25-10-2**	**Thu 25-11-06**	
19		Define Redemption Rules	2 days	Mon 25-11-0	Tue 25-11-04	9
20		Design UI Catalog	7 days	Wed 25-10-2	Thu 25-11-06	
21		**Backend & Rewards Database Design**	**10 days**	**Thu 25-11-06**	**Wed 25-11-19**	**19**
22		Design SQL Rewards Tables	2 days	Thu 25-11-06	Fri 25-11-07	
23		Implement Redemption Logic	7 days	Mon 25-11-1	Tue 25-11-18	22
24		Create RESTful API	3 days	Mon 25-11-1	Wed 25-11-19	
25		**Testing & QA**	**2 days**	**Thu 25-11-20**	**Fri 25-11-21**	**21**

Figure 6: Gantt Chart v3, Durations & Dependencies [5]



26	Perform Functional Tests	2 days	Thu 25-11-20	Fri 25-11-21	
27	Backend Core Completion	0 days	Thu 25-11-20	Thu 25-11-20	
28	Frontend Integration	0 days	Thu 25-11-20	Thu 25-11-20	
29	**Leaderboard Feature Development**	**17 days**	**Mon 25-11-0**	**Tue 25-11-25**	**9**
30	**Metric & Ranking Logic**	**7 days**	**Mon 25-11-0**	**Tue 25-11-11**	
31	Ranking & Tie-Breakers	1 day	Mon 25-11-0	Mon 25-11-03	
32	Develop Ranking Backend	6 days	Tue 25-11-04	Tue 25-11-11	31
33	**Frontend Leaderboard**	**8 days**	**Mon 25-11-1**	**Wed 25-11-19**	
34	Create Leaderboard UI	8 days	Mon 25-11-1	Wed 25-11-19	
35	Integrate Live Data Feed	6 days	Wed 25-11-1	Wed 25-11-19	32
36	**Testing & Optimization**	**4 days**	**Thu 25-11-20**	**Tue 25-11-25**	
37	Validate Leaderboard Accuracy	3 days	Thu 25-11-20	Mon 25-11-24	35
38	Optimize Leaderboard Performance	1 day	Tue 25-11-25	Tue 25-11-25	37
39	Module Testing & Validation Complete	0 days	Wed 25-11-2	Wed 25-11-26	
40	System Integration & Synchronization Verified	0 days	Fri 25-11-28	Fri 25-11-28	
41	Deployment Readiness	0 days	Mon 25-12-0	Mon 25-12-01	

Figure 7: Gantt Chart v3, Durations & Dependencies [5]



42		**System Integration & Deployment**	**5 days**	**Tue 25-11-25**	**Mon 25-12-01**	**6,17,29**
43		**Cross-Module Integration**	**3 days**	**Tue 25-11-25**	**Thu 25-11-27**	
44		Gamification Connection	2 days	Tue 25-11-25	Wed 25-11-26	
45		Verify Data Synchronization	1 day	Thu 25-11-27	Thu 25-11-27	44
46		**Development & Environment Setup**	**2 days**	**Fri 25-11-28**	**Mon 25-12-01**	**43**
47		Containerize Service via Docker	1 day	Fri 25-11-28	Fri 25-11-28	
48		Deploy Module to Production	1 day	Mon 25-12-0	Mon 25-12-01	47
49		Project Completion	0 days	Tue 25-12-02	Tue 25-12-02	

Figure 8: Gantt Chart v3, Durations & Dependencies [5]

13. Resource Assignment

Prepared by: Clayton Barnett

Date: 10/24/2025

To allocate resources for this project, we first need to be clear about the resources that we have available. We are assuming that we have a team of five general developers, one with some additional Project Manager responsibilities.

All tasks listed here refer to the updated finalized GANTT chart (version 3) with durations and dependencies, section 12 of this report.

Assigning People to Tasks Week-by-Week

WEEK 1
The project's first tasks (according to chronological order) include three broad business-level tasks ('Organization Alignment', 'Market Viability', and '3rd Party Cost Analysis'), 'System Rules' and 'SQL Schema', and 'Design UI Catalog'. These tasks can be split among the five members of the group. Because 'System Rules' and 'SQL Schema' are dependent on each other, it makes sense to give this task to one team member. 'Design UI Catalog' is the longest of these tasks.

WEEK 2
Four of the five team members will have finished their first tasks by the start of Week 2. The team member who did 'SQL Schema' can now start on 'Data Flow Diagram'. The three team members who did a business-level task can start on 'Project Architecture Development', 'Metric and Ranking Logic', and 'Define Redemption Rules', respectively. The 'Define Redemption Rules' team member can move to 'Design SQL Rewards Tables' in mid-week 2. The 'Metric and Ranking Logic' team member will be occupied with those two tasks until week 3. The team member who will finish the 'Data Flow Diagram' in mid-week 2 will complete the 'Requirements Finalization Task' and then join the 'Project Architecture Development' team.

WEEK 3
The team member who worked on ‘Design UI Catalog’ for the first two weeks will start on ‘Create Leaderboard UI’ in week 3. The ‘Project Architecture Development’ team will complete the ‘Database Schema & Data Flow Approval’ task at the start of Week 3, continue with ‘Project Architecture Development’ until mid-week 3, and then work on ‘RESTful Endpoints’ for the rest of week 3. The team member who did ‘Design SQL Rewards Tables’ will start on ‘Implement Redemption Logic’ at the start of week 3. The ‘Metric and Ranking Logic’ team member will finish those tasks mid-week 3 and then start on ‘Integrate Live Data Feed’.

WEEK 4
The two ‘Project Architecture Development’ team members will start on the ‘Create RESTful API’ and ‘Business Logic’ tasks, respectively, at the start of Week 4. The ‘Business Logic’ task member will then go immediately into the ‘Public Endpoints’ task. This will leave all team members free by Thursday of week 4. Three team members will then complete ‘Backend Core Completion’ and start on the ‘Perform Functional Tests’ task, while the other two will complete ‘Frontend Integration’ and then start on the ‘Validating Leaderboard Accuracy’ task.

WEEK 5
The ‘Validate Leaderboard Accuracy’ team will move to ‘Optimize Leaderboard Performance’, ‘Model Testing and Validation Complete’, and lastly ‘Containerize Service via Docker’ in week 5. The ‘Perform Functional Tests’ team will move to ‘Gamification Connection’, ‘Verify Data Synchronization’ and lastly ‘System Integration and Synchronization Verified’ in Week 5.

WEEK 6
The entire team will work on ‘Deployment Readiness’ at the start of Week 6 before moving to ‘Deploy Module to Production’, and lastly, the ‘Project Completion’ milestone.

Hours-per-task Table
We assumed that devs worked 7 hours per day, 5 days a week. Some weeks, a dev (particularly the PM) does not spend 35 hours on tasks. This extra time could be used for additional documentation, meetings, or, in the case of the PM, project-management tasks.

Task	Week	Dev A	Dev B	Dev C	Dev D	P M	Notes
Organization Alignment	1	0	0	0	0	28	Business-level task
Market Viability	1	0	35	0	0	0	Business-level task

3rd Party Cost Analysis	1	0	0	35	0	0	Business-level task
System Rules	1	0	0	0	7	0	Predecessor to SQL Schema
SQL Schema	1	0	0	0	28	0	Preceded System Rules
Design UI Catalog	1–2	49	0	0	0	0	Longest early task
Data Flow Diagram	2	0	0	0	7	0	Follows SQL Schema
Project Architecture Development	2–3	0	52	0	38	0	Dev D joins Tuesday of Week 2
Ranking and Tie-Breakers	2	0	0	7	0	0	First task of ‘Metric and Ranking Logic’
Develop Ranking Backend	2-3	0	0	42	0	0	Runs into Week 3
Define Redemption Rules	2	0	0	0	0	14	Moves to SQL Rewards mid-week
Design SQL Rewards Tables	2	0	0	0	0	14	Follows Redemption Rules
Requirements Finalization	2	0	0	0	7	0	Done after DFD
Database Schema & Data Flow Approval	3	0	4	0	4	0	Early Week 3
RESTful Endpoints	3	0	14	0	14	0	After Architecture task
Create Leaderboard UI	3	56	0	0	0	0	From UI Catalog dev
Redemption Logic	3-4	0	0	0	0	49	Follows SQL Rewards Tables
Integrate Live Data Feed	3	0	0	42	0	0	After Metric & Ranking Logic
Create a RESTful API	4	0	21	0	0	0	Week 4 start

Business Logic	4	0	0	0	10	0	Then Public Endpoints
Public Endpoints	4	0	0	0	4	0	After Business Logic
Backend Core Completion	4	0	4	0	4	4	Shared by 3 members
Perform Functional Tests	4	0	10	0	10	10	
Frontend Integration	4	4	0	4	0	0	Done by UI-related devs
Validate Leaderboard Accuracy	4–5	17	0	17	0	0	Follows Frontend Integration
Optimize Leaderboard Performance	5	7	0	7	0	0	Week 5 starts
Model Testing & Validation Complete	5	7	0	7	0	0	After Optimization
Containerize Service via Docker	5	7	0	7	0	0	Final Week 5 task
Gamification Connection	5	0	14	0	14	14	From the Functional Tests team
Verify Data Synchronization	5	0	7	0	7	7	Mid-Week 5
System Integration & Synchronization Verified	5	0	7	0	7	7	End of Week 5
Deployment Readiness	6	4	4	4	4	4	All-team task
Deploy Module to Production	6	3	3	3	3	3	All-team task
Project Completion	6	7	7	7	7	7	Milestone closure

Table 12: Hours-per-task Breakdown

References

[1] J. Li, *"1.1 Financial Analysis,"* CPS 714 Software Project Management, Microsoft Excel file, Oct. 12, 2025.

[2] J. Li, *"2.4 Gantt Chart,"* CPS 714 Software Project Management, Microsoft Project file, Oct. 20, 2025.

[3] J. Li, *"3.3 Gantt Chart Changed,"* CPS 714 Software Project Management, Microsoft Project file, Oct. 23, 2025.

[4] J. Li, *"3.3 AON Network Diagram,"* CPS 714 Software Project Management, Portable Network Graphics (PNG) file, Oct. 23, 2025.

[5] J. Li, *"3.4 Gantt Chart Durations & Dependencies,"* CPS 714 Software Project Management, Microsoft Project file, Oct. 24, 2025.